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Geospatial Database Design for implementing enhanced Local Governance with spatial analysis: A Case study of Manesar area – India

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Abstract

Multi-agency investments require local coordination, planning, and implementation. GIS allows us to analyze and comprehend real-world processes by constructing and applying manipulation, analytical criteria, and models for daily decision making. This requires a geospatial database with surveyed land, assets, hydro, etc. Survey of India (SOI) is publishing a topographical map of India with significant cities, areas of interest, and a geodetic benchmark. Different states in India have set up a complete geographic database with land parcels, asset mapping, and administrative borders usually based on the national land record modernization program survey. Many state and local governments also have geospatial databases. Existing geospatial database designs are based on individual requirements that have generated a large volume of duplicate, unformatted, unstructured data. These datasets lacked coordination or logical intercorrelation with other departments' datasets. It served a single department's needs. This causes redundant data, low quality, and a lack of data understanding. The problem's appropriate representation is key to solving it. The Study creates a Common Geo Database for the region to improve collaboration and working. After consulting civil, electrical, and planning engineers, a common database is created. GIS adoption has improved day-to-day decision-making.

1 Introduction

Rapidly increasing population and economic growth have fostered urbanization, one of the biggest challenges developing countries like India face, putting pressure on the whole governance (Manjula et al., 2012). The best possible way to manage such a huge landmass and population is to decentralize the governance. While the constitution was prepared, local government subjects were assigned to the states, and it was one of the provisions in the directive principle of state policy (Hazra et al., 2013). India has adopted decentralization in a big way by having the Panchayati raj established. This was done after the 73rd amendment in the constitution resulting in uniform three-tier panchayat raj established in all states. From bottom to top the three-tier structure includes gram panchayat, Mandal, and then Zila panchayat (Hazra et al., 2013). Local Governance enables the government to have the advantage of local knowledge and thereby make decisions early and easily. There are many factors associated with the low efficiency of local governance but one of the main factors responsible is the non-availability of spatial as well as linked real-time non-spatial information which can be used by all (Anuj et al., 2014; Manoj et al., 2016).

The information and the services needed to deal with it can be readily transported by electronic means and geospatial information is increasingly combined with emerging sensor, visualization, and interaction technologies (Manoj et al., 2016). Further, it is often seen that the use and application of Geospatial information technology are not effectively used due to distinct reasons such as lack of updated information, missing information on terrain, water network, inventory management, drainage management data models, etc. There is a great need to apply Geospatial technology for use at the local level to improve the development work done by the government to enhance the lifestyle of people (Anuj et al., 2014). In order to alleviate this problem, it is of utmost importance that the decision-makers at the local level, such as the gram panchayat, District administration, and Municipal corporations, are fully

equipped with the geospatial information that is required for the activities that are associated with development. A Geographical Information System or GIS is a computer-based system, which includes capturing, storing, analyzing, and managing all kinds of spatial data (Varma et al., 2014). Geospatial data with updated real-time information can be a future of fastening decision-making for sustainable local self-governance. Different departments across the local government set up their own geodatabase for their day-to-day analysis and decision-making (Kumar et al., 2010). Examples of successful systems include those for the collection of municipal taxes and national land records. However, the connection between the geodatabases that are available on the local, state, and national levels is not nearly as efficient as it could be. Because of this, there are challenges involved in putting the project that is sponsored by the state or the center into action at the local level. After decentralization, local authorities handle controlling, checking, and implementing the various projects using geospatial datasets (Balwan et al., 2018). The benefit of having common geospatial datasets for all levels of governance has numerous benefits in decision making. Our study describes the design of a common geospatial database that can successfully bring geographical information for decision-making by stakeholders at all three levels of governance. The most common problems faced in the developing countries like India include the lack of coordination among agencies and lack of onsite data available for the officers. Some of the most common issues are depicted in Figure.1.

The purpose of this research is to identify the many kinds of geographical data necessary for local governance in the Manesar area, so that they can be linked with a shared database those various stakeholders can use to make decisions more effectively in their day-to-day operations. By maintaining a current geospatial database, all levels of government and municipalities will have equal access to the data they need to do their jobs effectively and efficiently, eliminating potential for prejudice.

The availability of a common geospatial database and user-friendly layout to retrieve geographic information is a key towards its applicability and acceptance by end users. Visual representation of the data in vector and raster formats using GIS software's has been considered as a key crucial factor in many studies (Manjula et al., 2012). This emphasizes the visual interpretation of spatial data for decision-making. GIS database is a combination of various attribute information in tabular form that is stored in a structured format (Karabegovic et al., 2018). The database management system of GIS software, such as Arc GIS from ESRI, may make use of these tabled connections between data through a number of different types of structural queries. The geospatial database is like that of other conventional databases however it is differentiated from each other based on its ability to store, manipulate, retrieve, and display spatial data (A.Musa et.al). Database management systems (DBMS) on the other hand are specially designed application software packages that interact with the user, other applications, and the database itself to capture, query, and analyze data (Wikipedia, 2013). GIS-based Spatio-temporal analysis done by Suzanchi K et.al has analyzed the changing land use pattern of the area of study.

Available studies concentrate on developing database models for particular applications, like asset mapping, landslide analysis, and Land use Landcover analysis. Each of these databases can be used independently for research into its own specialty area. However, there is little investigation into how to go

about making a central database that can be accessed by all branches of government to facilitate decentralized administration. Users were consulted during the current research's preparation of a standardized geospatial database design. Its use in an online app and its subsequent conclusion have both been presented in great detail.

2 Pilot Study Area

The selection of the study area (Figure.2) was done considering the existence of three tiers in the local governance i.e., Gram panchayat, block, and then district. In the study area, there are 35 villages under block/Tehsil Manesar which lies under district Gurugram in the state of Haryana. Manesar (Lat 28.354285°, Long 76.939820°) is one of the five Tehsil of Gurugram district and a fast-growing industrial township. As per the census result of 2011 the total population of the Manesar tehsil/block is 116606 people.

3 Methodology

In order to effectively display, retrieve, and manipulate spatial and non-spatial information, a Geographical Information System relies on a robust data model. Therefore, a well-defined data model is required for geodatabase design, and a model cannot be created without gathering requirements from many departments. Through unstructured interviews with members of panchayats, municipal councils, and government officials at the local level, we were able to learn about the perspectives of those most directly involved in the issues we were investigating. The discussion's stated objective was to get insight into the current use of GIS technology in operational and decision-making contexts. Users' requirements are being evaluated on a panchayat-wide and government-wide scale. The information available on the different government websites that can be referred to separately is not effective in decision making (Jain et al., 2016). Government departments are using different databases which are not linked with each other, at times we find Geospatial database is old and there is no real-time Update. Available geospatial maps or web-based applications are not intelligent enough for day-to-day usage and decision-making (Kaushik et al., 2017).

The latest available Landsat Satellite Imagery (year 2021) of the study area was downloaded from the USGS (United States Geological Survey) site for preparing the base map. The downloaded image was multispectral and hence for preparation of vector base map, RGB bands 4, 3, and 2 were used. Here all the defined features were captured and then classified in their sub classes such as water bodies were classified as Drainage, ditch, lake, river, seasonal for better analysis as per requirement (Amini et al., 2016). For the GIS project, database development is one of the most important activities. Data modelling, or what is commonly called database design, is a beginning step in database development (Ponjavic et al., 2019). Data modeling is the first step in the database design process, and the database software chosen for implementation is the second. Bentley Systems MicroStation version 08.05.02.35, Geographics 2004 edition, and Microsoft Access version 2016 were used to generate vector data in our study. The CAD (Computer Aided Design) environment data was captured using Bentley MicroStation,

and the attributes were stored in an MS Access database. Orthophoto was used as a raster to manually digitize the features and create a map of the world. The 2021 Landsat satellite images were used to create the orthophotos. MicroStation was used to design the features in Table.1, and digitization was finalized with help from georeferenced satellite imagery. Figure.3 is a quick look at some key land-use characteristics of the research area.

4 Conceptualisation Of Relational Database Management System (Rdbms)

Design of database model is a process of conceptual representation of the data structures in a database (Carrion et al.,2009). Whereas data structures form objects of data, relationships between data objects and rules which regulate operations on the objects (Zhouetal et al., 2020). The Relational model is simple to implement and efficient in its results. It enables the use of a high-level query language, Structured Query Language (SQL), which can be employed to forego laborious manual database browsing. It is easy to increase or decrease the number of records, rows, and columns in a relational database without modifying any of the dependent applications. Data models are the core of any GIS which gives a set of constructs for describing and representing selected aspects of the real world in a computer (Longley et al., 2005). By preventing data from being kept in several locations, RDBMS guarantees both the accuracy and integrity of its data. Using Figure.4, we can see the overall process for developing our RDBMS.

Research objectives

This task's goal is to create a reliable database design for storing the spatial data and the non-spatial information that goes along with it. The purpose of design is to facilitate the use of a shared geospatial database for a wide range of planning and decision-making purposes across a variety of stakeholder groups.

Requirement Understanding *(Brain Storming)*

Our consultations with numerous government entities informed the database's layout. A brainstorming session with various government agencies is included. Repair and maintenance of road assets such Kerbs, traffic islands, maintenance holes, ditches, electric poles, etc. were identified as major problems. Major difficulties for engineers were waterlogging and drainage, encroachment, and unlawful building. When analyzing a problem that requires collaboration between government agencies, those agencies were employing incompatible sets of spatial data. Since many parties use their own data to satisfy their needs, coordination among agencies has been extremely limited.

This slowed down the process of making decisions to address the problems that were uncovered. At the end of the brainstorming session, the most often occurring characteristics had been identified and separated into their respective categories, attributes, and subattributes as depicted in Table.1.

Table.1. Feature and category of the prepared base map

GMLTableName	ManesarTableName	Category
Electric Pylon	a403_Electric_Pylon	Technical Installations
Road Center Line	a301_RoadCentreLine	Traffic
Building	a101_Building	Buildings
Urban High Density	a201_UrbanHighDensity	Built-up Area
Industrial Low Density	a202_IndustrialLowDensity	Built-up Area
High Tension Wire	a405_HighTensionWire	Technical Installations
Lake	a601_Lake	Water
Water Way	a602_WaterWay	Water
Railway	a303_Railway	Traffic
Rocky Area	a505_RockyArea	Vegetation
Parking Area	a306_ParkingArea	Traffic
Settlement Village	a205_SettlementVillage	Built-up Area
Under_Construction	a207_Under_Construction	Built-up Area
Water Treatment Plant	a408_WaterTreatmentPlant	Technical Installations
Electric Substation	a415_ElectricSubstation	Technical Installations

Identification of data types

After compiling all the data, the recognized characteristics were separated into their own groups, as shown in Fig. 5. It was then determined whether the feature was a point, line, or polygon based on its location. Attribute information requirements were later set for each field, including integer, Boolean, text, Date, etc. The columns containing Feature names have been given unique primary keys.

Creation of Relationships among tables

Relationships between tables are the most important aspect of the architecture, as this is where the power of a relational database is realized. To facilitate the development of a unified database, we have compiled a set of metadata describing each characteristic used in our research. One-to-one partnerships are the simplest, but there are also many-to-many and many-to-many relationships. Figure.6 shows some of the most important connections in our data set.

Refinement and Normalization of Design

The process of standardizing the design went through multiple versions. Categories and types of features are separated into separate tables. Databases that follow all normalization principles are guaranteed to

be well-structured. There was a single value in each cell of the first normal form, and then there was a second normal form, a third normal form, and a higher-order normal form. The honesty of the database has been tested by applying consistency standards.

This included the following checks:

- Primary Key does not have any NULL Values
- Each foreign key value matched to the primary key value
- Column Indexing was carried out to ease the data searching and retrieval. Different queries were defined to cater to the linkages and retrieval of data

Creation of Graphical user interface

A Graphical user interface was developed between Bentley's MicroStation and Database MS Access which helped in viewing and updating attributes of the features in real-time (Kaushik et al, 2017). The tool provides standard attribute assignment functionality for element capturing and editing of attributes. The primary functionality in the Attribute Assign tool is:

- Attribution during element capture (main purpose)
- Edit of attributes at existing elements
- Info on attributes at existing elements
- Update selected attributes at multiple elements
- Feature Code change with the handle of corresponding attribute linkage/table
- Querying database for elements in dgn-file, inside the fence, or displayed levels
- Validation of linkage at elements
- Validation of attribute values according to relationship rules

Attribute Assign works with two primary dialogs

- Feature Manager, i.e., alternative to Geographics Feature Manager, for selecting the Feature Code
- Attribute Assign the main dialog for attribution control

The main Attribute Assign (Figure.7.) dialog has a small menu providing access to configuration and some more specialized functions. For special purposes will be activated extra dialogs:

- AttrAssign: Model select
- AttrAssign: Info
- AttrAssign: Query
- AttrAssign: Update Multiple
- AttrAssign: Button Pad

Retrieval of Data through SQL queries

MicroStation's graphical user interface makes it easy for users to construct queries tailored to their needs and gain access to the corresponding data. MicroStation was integrated with the geospatial database that was built. Different line styles and colors have been used to make the characteristics and attributes more readily distinguishable. Several different kinds of SQL-based queries were executed as part of the testing procedure to gauge how well the database design met the criteria.

Attribute query and returned value for Polygon Features

The building feature has been plotted has closed Polygons. The Type of Buildings has been classified as Building (In red Colour) and Greenhouse (in Green Colour). Queries for retrieval of Building -Building is

Select *from a101_building where typeofbuildingid = 1 and mslink =

Whereas for the retrieval of Building-Greenhouse is

Select *from a101_building where typeofbuildingid = 3 and mslink =

a101_building is the building table and typeofbuilding is the attribute table

The query was returned with the display of target spatial data with its attribute information as shown in Figure.8 and Figure.9.

Attribute query and returned value for Line Features

Figure.10. and 11 shows the National Highway NH48 from Delhi to Jaipur. There are two attributes assigned to the Road center line. One gives the Type of Road, which in this case is a Highway and the other shares the Surface type of the road which in this case is “Consolidated” or paved. The queries for both the Attributes are

Select *from a301_roadcenterline where typeoftrafficid = 1 and mslink = and

Select *from a301_roadcenterline where surfaceid = 1 and mslink =

Typeoftrafficid = 1 denotes Highway and surfaced = 1 denotes the paved/consolidated road

Here, the extracted feature is the Road center line with the attribute Expressway. It is called the KMP Expressway and joins the Kundli, Manesar and palwal districts bypassing the capital Delhi. The query is

Select *from a301_roadcenterline where typeoftrafficid = 2 and mslink =

Where typeoftrafficid = 2 denotes the Expressway in yellow color

Figure.12. stands for the roads around the intersection where the KMP expressway and NH48 intersect with each other. The cyan colour depicts the NH48, and yellow colour depicts the KMP expressway.

However, the main aim of the above is to extract the other “All traffic” type of Road and the road surfaces with “unconsolidated” or unpaved roads. The green colour dotted lines depict the unpaved roads.

Select *from a301_roadcenterline where surfaceid = 2 and mslink =

Where surfaceid = 2 denotes the roads as “unconsolidated” or unpaved

Figure.13. depicts the extraction of Railway line and its attributes as “Type of Track.”

This is the Rewari line going from Delhi to Rewari and is both a passenger and freight corridor. Following queries can enable the access of the Rail lines

Select * from a303_railway where typeoftrafficid = 1 and mslink =

denotes the Main track used by the railways

Select * from a303_railway where typeoftrafficid = 2 and mslink =

denotes the other track in station

Select * from a303_railway where typeoftrafficid = 1 and mslink =

denotes the siding track wherein the dead end of the rail line is there

5 Result, Recommendation And Conclusion

The municipal, electrical, and transport departments in the Manesar region may now use the same comprehensive Geospatial information. Our method for dealing with shared geospatial information centers on the development of a database, a crucial part of any GIS. Through the use of MicroStation and the Microsoft database, a solid database architecture was developed, and a base map with topographical elements was made. Using these three GIS components, we can now pose structural attribute queries for requirement analysis (Hardware, software and Geospatial data).

Buildings, traffic, built-up area, technological installations, water, vegetation, and their associated qualities, and other aspects are mapped in relation to the entire area. Its unique type IDs and surface IDs were used to map a total of 27 key properties. The many feature types and attributes are indicated by their own unique hues. It's easy to tell the difference between joint surfaces (solid) and unconsolidated ones (dotted) because to the visual cues used to represent them. Topology, data editing, and attribute queries in the GIS database are all fully functional, allowing for reliable, day-to-day decision making. Specifically crafted SQL queries were used to emphasize and validate the database. Data was used by field engineers, town planners, and related offices for attribute-based inquiry during testing and deployment. This dataset is special since it provides a unified location for all data, which can be refreshed at any moment within a predetermined time frame. Buildings, traffic, built-up area, technological

installations, water, and vegetation are all examples of geographical data that may be retrieved with a standard SQL query.

Recommendation

An online intelligent geospatial database that links panchayats to blocks and ultimately to districts was identified as a necessary improvement. As seen in Figure.14, the approach should consist of three distinct parts.

Segment one (Temporal Data)

All relevant historical information is compiled into a database for further use in GIS-based analysis. Orthophotos, historical vector data related to census records, weather records, crop patterns, etc. may all be a part of this.

Segment Two (Control Segment)

This section has the potential to aid in the real-time insertion of geographic and non-spatial data into the database. The map and its associated attributes may be refreshed to reflect recent events or developments. In this case, the operator can also play the role of moderator, approving data and making changes to the dataset in response to requests from the end user or a third party.

Segment three (Web Based user interface)

In theory, this part of the site might be accessible to visitors who wish to glean data pertinent to their specific interests. In this context, you may be able to choose for real-time updates, such as the most current crop pattern in each village's panchayat. You can use this data to see what kinds of grains and veggies are readily available in the area.

The inability of several government entities to collaborate productively results in additional expenses for the general population. Some examples of this include multiple organizations excavating the same road twice, erroneous taxing, and encroachments that have not been notified. According to the findings of our investigation, a GIS-based solution is available in the form of a shared geographic database that numerous government agencies are able to access concurrently.

It was discovered that there is a requirement for a web-based shared intelligent geospatial database that is able to connect panchayats to blocks, and subsequently blocks to districts. As part of our research, we have compiled a Geodatabase that can help all the local governing bodies work together to make decisions that are consistent with reason. As a result of the study, a vector map of the study region will be created, and a web-based user interface will be suggested. This interface will be usable by end users at all levels of a three-tier Panchayat system. By analyzing and gaining access to the most recent information, this will be of assistance to the relevant stakeholders in the process of making day-to-day decisions. This

will also help in effective functioning and speedy decision making, eliminating the lag time and saving both time and money in the process.

The database has the potential to be expanded to include other traits and attributes that are pertinent to our research area. After going through our research, we have come to the conclusion that the Government of India should establish a high-level team comprised of experts from a variety of fields in order to investigate and develop a common geospatial database specification for the entirety of the nation. This specification should be utilized by all branches of the government as well as the private sector.

6 Conclusion

Spatial Data is driving today's global decision-making. Technological advancements have improved analysis. The spatial data can be updated regularly and hosted on a web server. Lack of a standard geo spatial database can be solved, but it will need a lot of will and collaboration from many government entities. GIS web-based spatial analysis is aided by a consistent database design. Sql-based queries allow stakeholders to quickly apply spatial analysis results in decision-making. This paper provides a viable way to create a common geospatial database for local government stakeholders. During the study, a web-based application prototype that links the shared geographic database for geographical query was given, along with implementation suggestions. A spatial database can be a compromise if its spatial data processing skills and industry requirements are ignored. Our research demands project-oriented creation and customization of spatially responsive SQL for a clear user interface. Innovative spatial analysis capabilities were once confined to desktop GIS. Our research combined search, change, analyze, manage, and save data into a single web browser interface. This is effective for applications and development.

Declarations

competing interest

It is stated that the authors have no competing financial interests or personal relationships that may have influenced their work.

AUTHORS' CONTRIBUTIONS

The authors confirm contribution to the paper as follows: Author, study conception and design: **Manuj Dev.** Author analysis and interpretation of results, **Vanama Venkata Sai Krishna**. Author analysis and interpretation of results: **Deepak Kumar**. Author; analysis and interpretation of results: **Ashutosh Kainthola.** Author; draft manuscript preparation; analysis and interpretation of results: **Manas Kumar Jha**. All authors reviewed the results and approved the final version of the manuscript.

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AVAILABILITY OF DATA AND MATERIALS

The authors confirm that the data supporting the findings of this study are available within the article [and/or] its supplementary materials. It can be produced if required during the review of the paper from corresponding author.

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Figures



Figure 1

The most common issues where geospatial data can be used for decision making

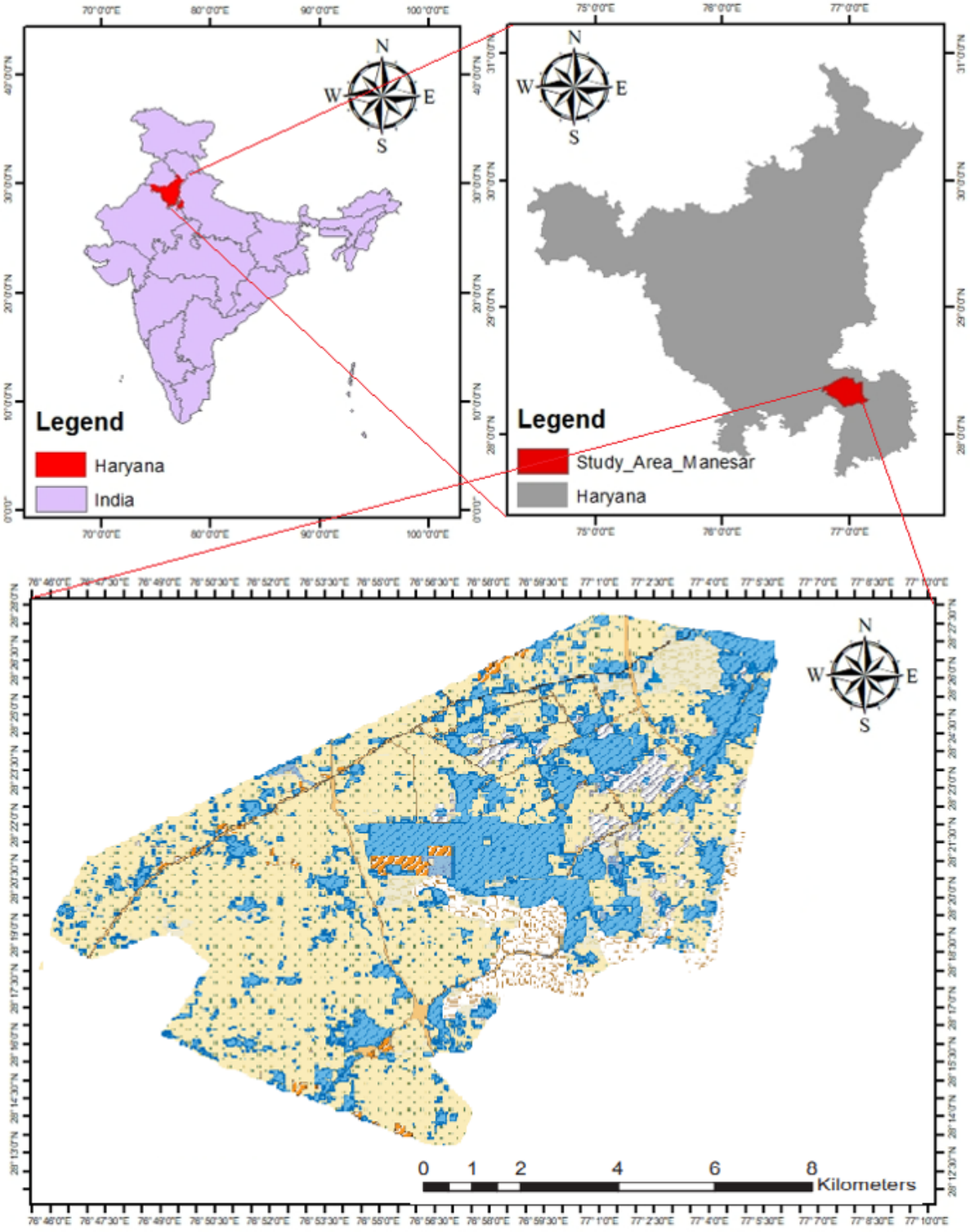


Figure 2

Study area

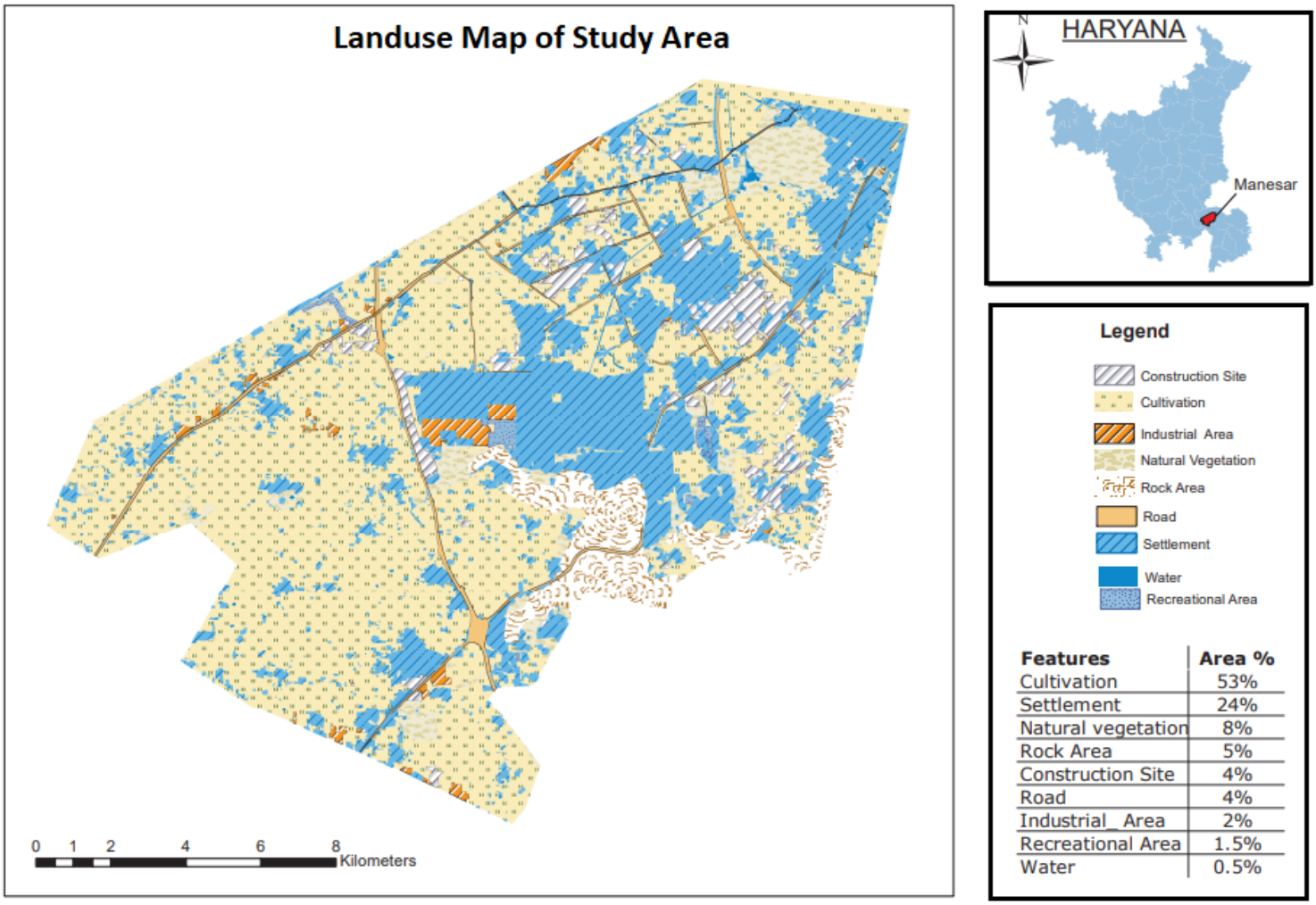


Figure 3

land use map of study area

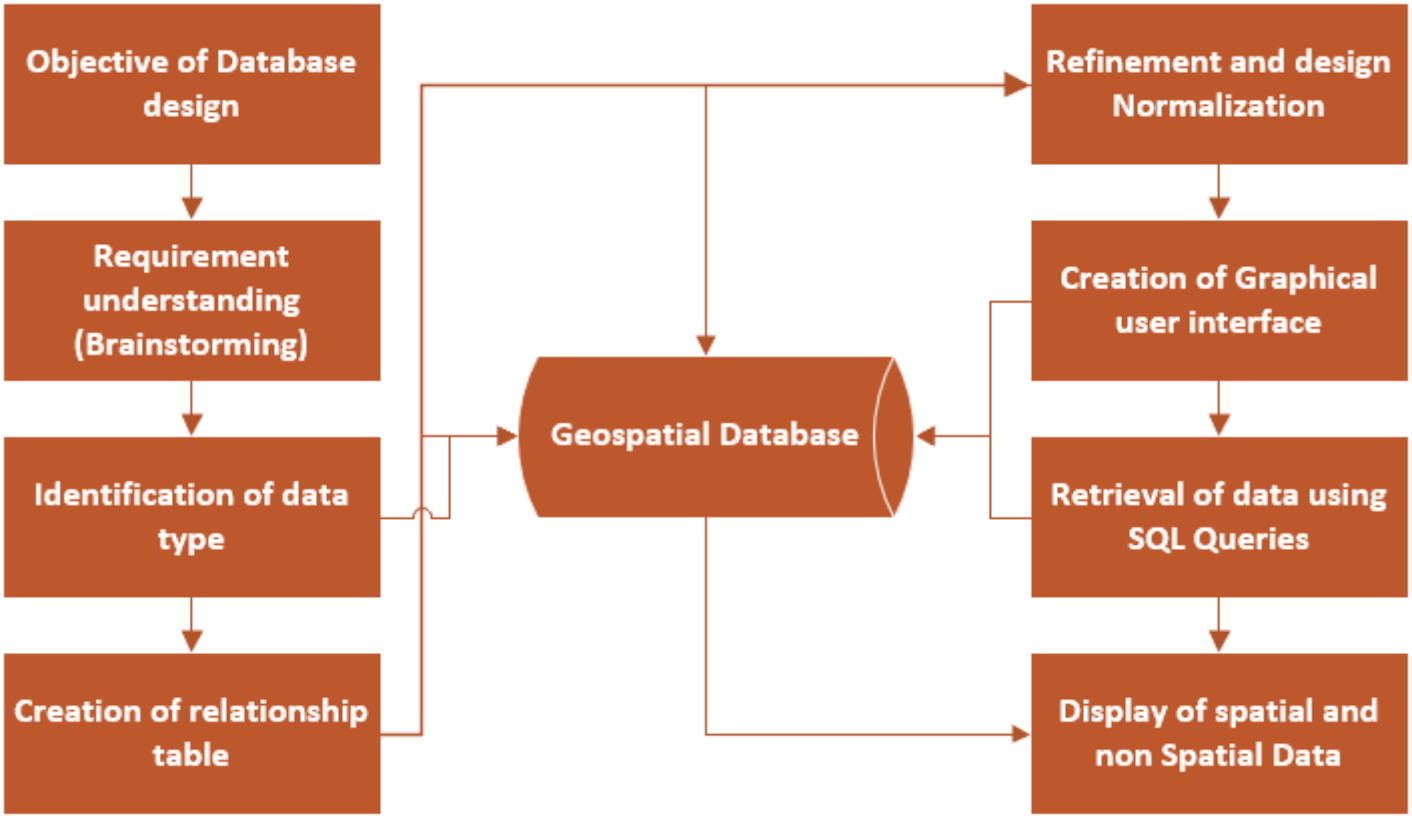


Figure 4

Approach for Geo-Database Creation

table0	index0	table1	index1	value1	descr	picklist_dsf
a101_Building	TypeOfBuildingID	PL_101_TypeOfBuilding	mslink	Value_DK		1
a000_default	MetadataId	Metadata	mslink	mslink		0
a301_RoadCenterLine	TypeOfTrafficID	PL_301_TypeOfTraffic	mslink	Value_DK		1
a301_RoadCenterLine	SurfaceID	PL_301_Surface	mslink	Value_DK		1
a301_RoadCenterLine	RoundaboutID	PL_000_Boolean	mslink	Value_DK	Description	1
a303_Railway	MetadataId	Metadata	mslink	mslink		0
a303_Railway	TypeOfTrackID	PL_303_TypeOfTrack	mslink	Value_DK		1
a303_Railway	LevelID	PL_303_Level	mslink	Value_DK		1
a304_RoadEdge	TypeID	PL_304_Type	mslink	Value_DK		1
a304_RoadEdge	SurfaceID	PL_304_Surface	mslink	Value_DK		1
a307_TrafficFence	TypeOfFenceID	PL_307_TypeOfFence	mslink	Value_DK		1
a401_TechnicalArea	TypeOfAreaID	PL_401_TypeOfArea	mslink	Value_DK		1
a402_BasinReservoir	TypeOfBasinID	PL_402_TypeOfBasin	mslink	Value_DK		1
a405_HighTensionLine	TensionID	PL_405_Tension	mslink	Value_DK		1
a410_Telemast	TypeOfMediumID	PL_410_TypeOfMedium	mslink	Value_DK		1
a602_WaterWay	NetworkID	PL_000_Boolean	mslink	Value_DK	Description	1
a602_WaterWay	VisibleID	PL_000_Boolean	mslink	Value_DK	Description	1
a605_Harbour	TypeOfHarbourID	PL_605_TypeOfHarbour	mslink	Value_DK		1
a607_Jetty	TypeOfJettyID	PL_607_TypeOfJetty	mslink	Value_DK		1
a607_Jetty	LocationID	PL_607_Location	mslink	Value_DK		1
a701_City	TypeOfCityID	PL_701_TypeOfCity	mslink	Value_DK		1
a801_AreaPolygon	TypeID	PL_801_Type	mslink	Value_DK	Description	1
a506_RawMaterial	TypeOfRawMaterialID	PL_506_TypeOfRawMaterial	mslink	Value_DK		1
a510_LandUseBoundary	TypeOfLandUseBoundaryID	PL_510_TypeOfLandUseBoundary	mslink	Value_DK		1
a511_Slope	TypeOfSlopeID	PL_511_TypeOfSlope	mslink	Value_DK		1
Metadata	hovedopl_id	hovedoplysninger	mslink	navn_leverandoer		0
a203_IndustrialHighDensity	MetadataId	Metadata	mslink	mslink		0
a201_UrbanHighDensity	MetadataId	Metadata	mslink	mslink		0
a202_IndustrialLowDensity	MetadataId	Metadata	mslink	mslink		0
a415_ElectricSubstation	MetadataId	Metadata	mslink	mslink		0

Figure 5

Data organized in the Tables

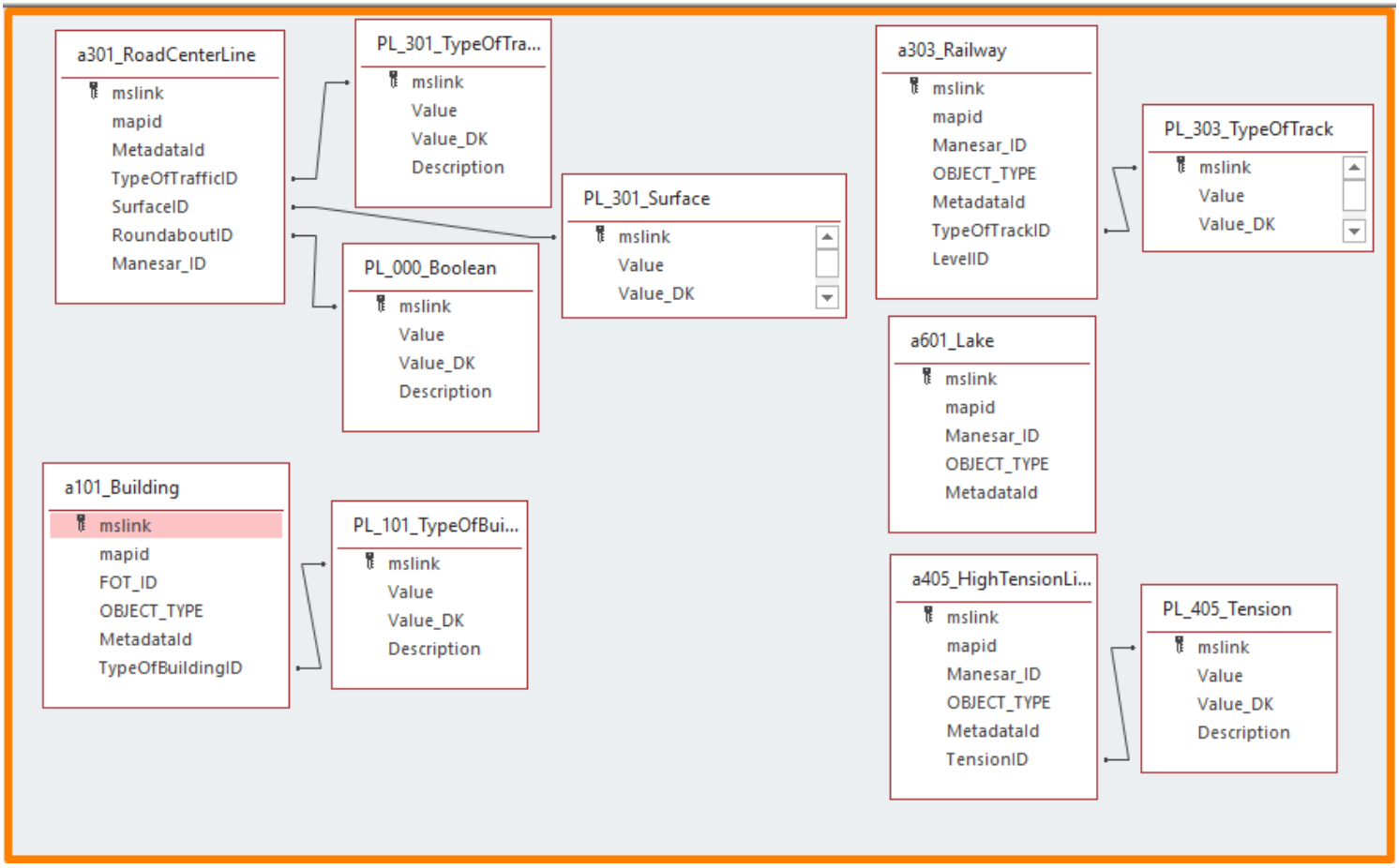


Figure 6

Relationship Table

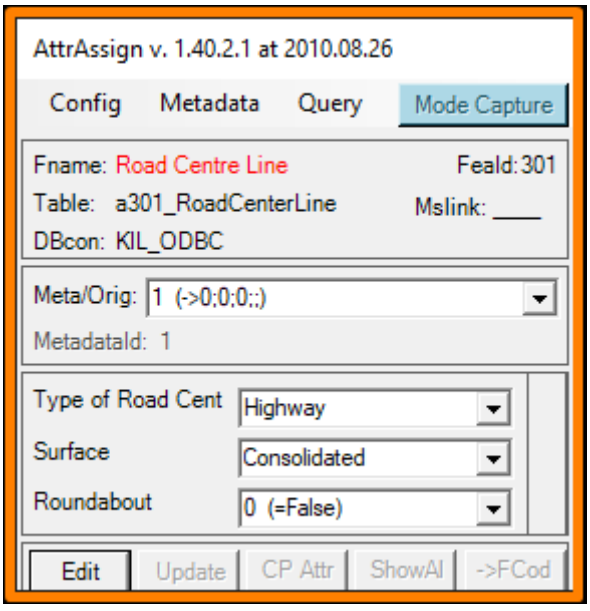


Figure 7

Graphical User Interface



Figure 8

Query results with building polygon and its attribute information

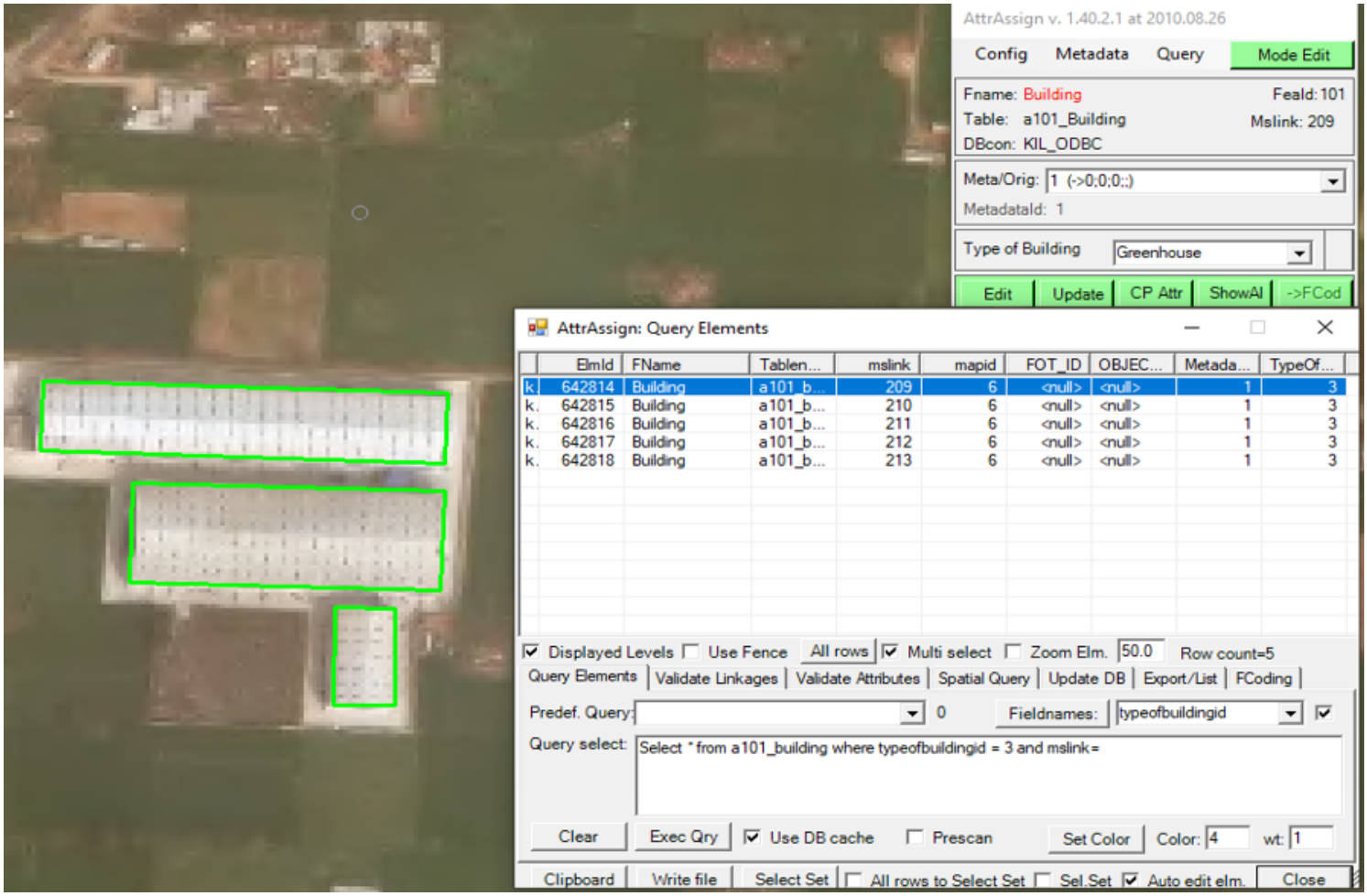


Figure 9

Query results with green house polygon and its attribute information

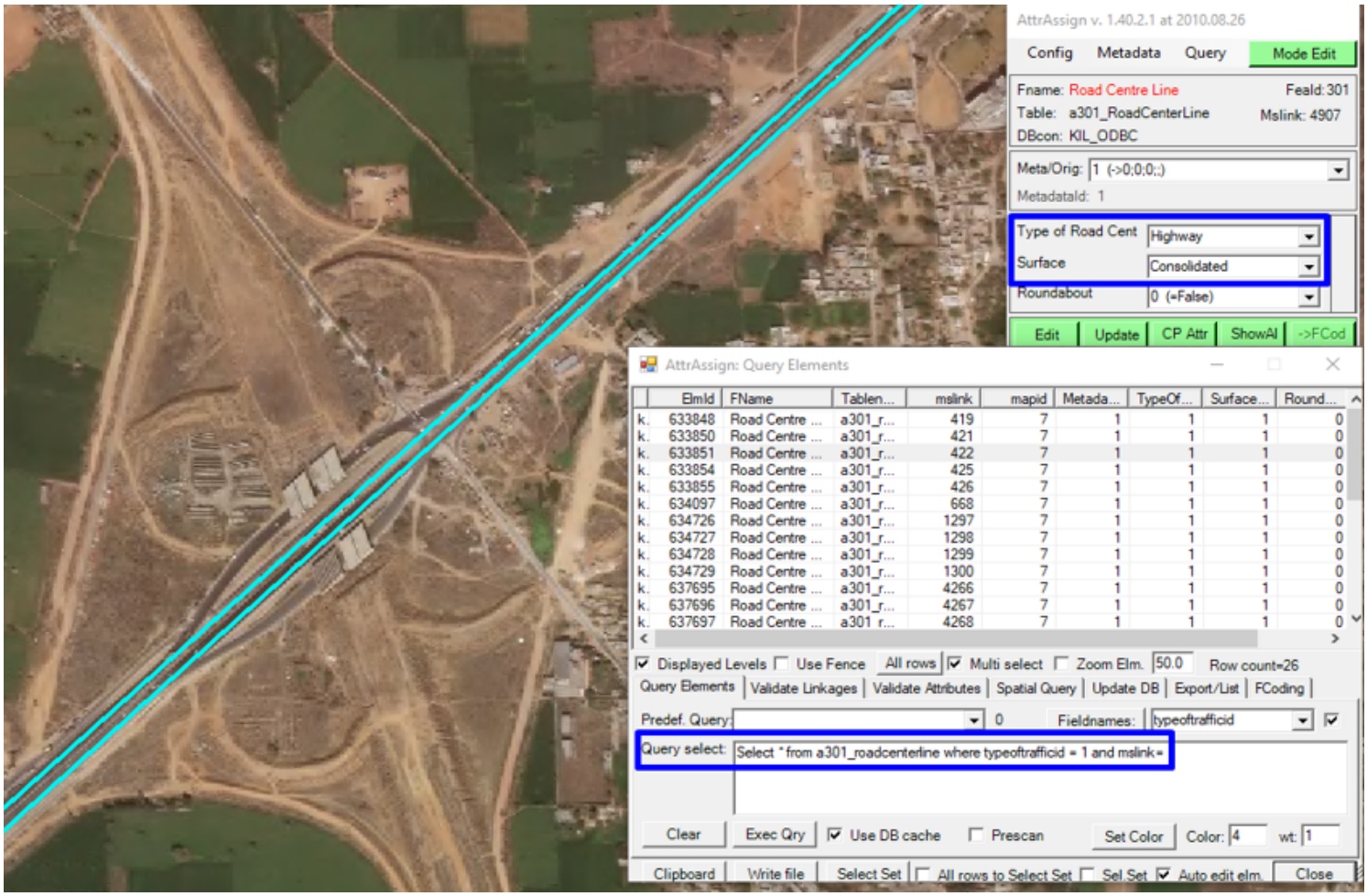


Figure 10

result of attribute query for line feature – Road

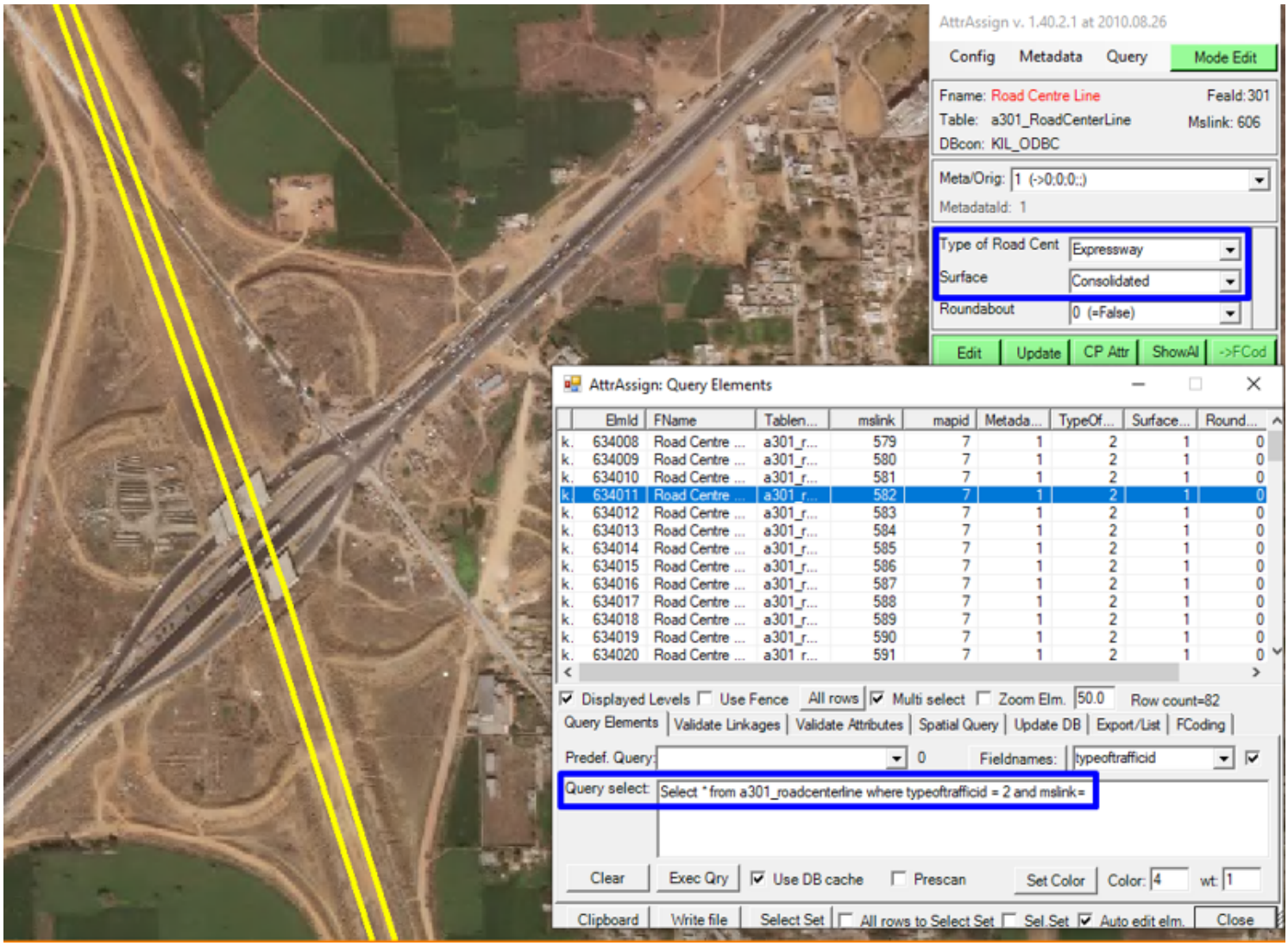


Figure 11

result of attribute query for line feature – Expressway

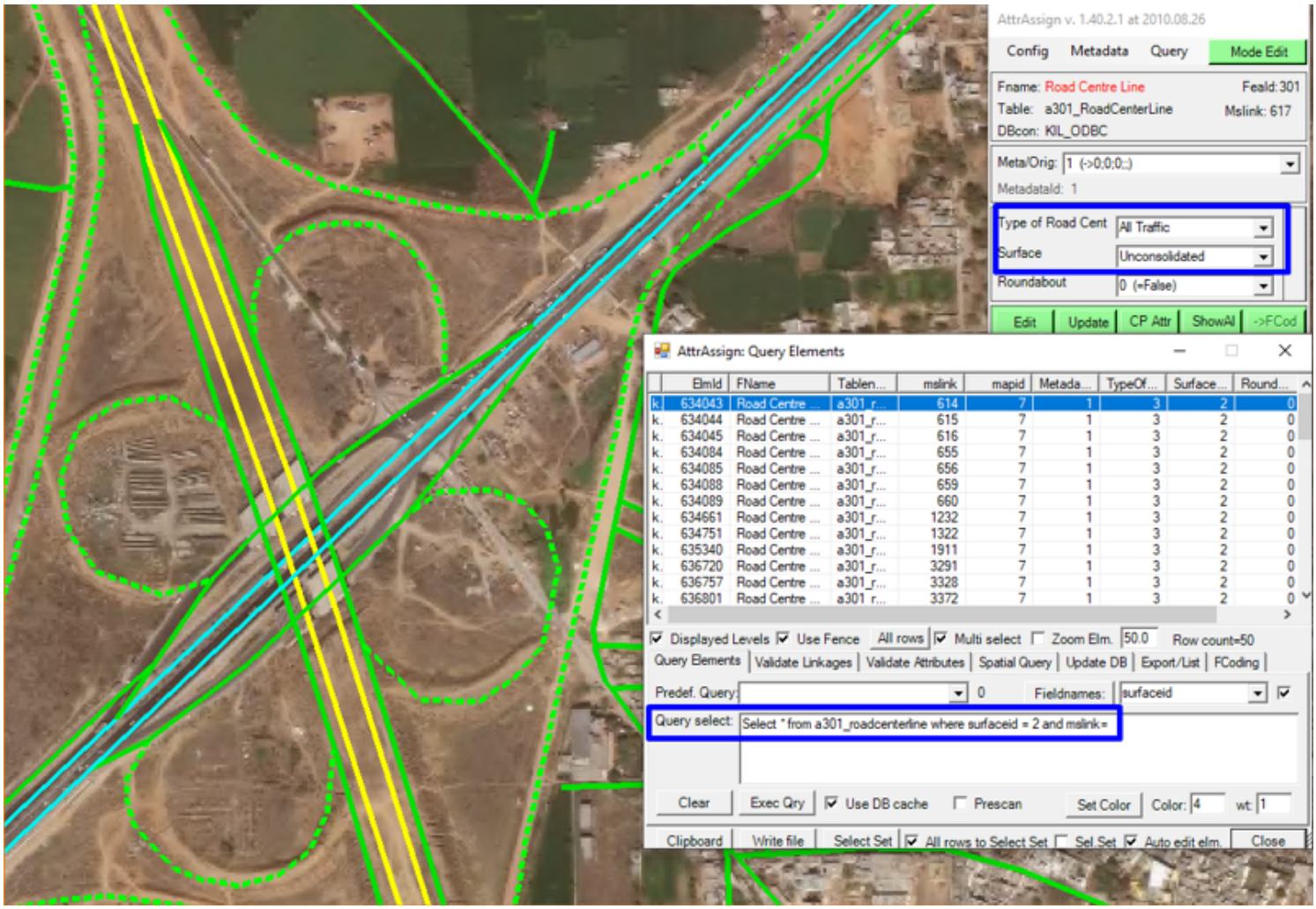


Figure 12

result of attribute query for line feature - Expressway



Figure 13

result of attribute query for line feature – Railway track

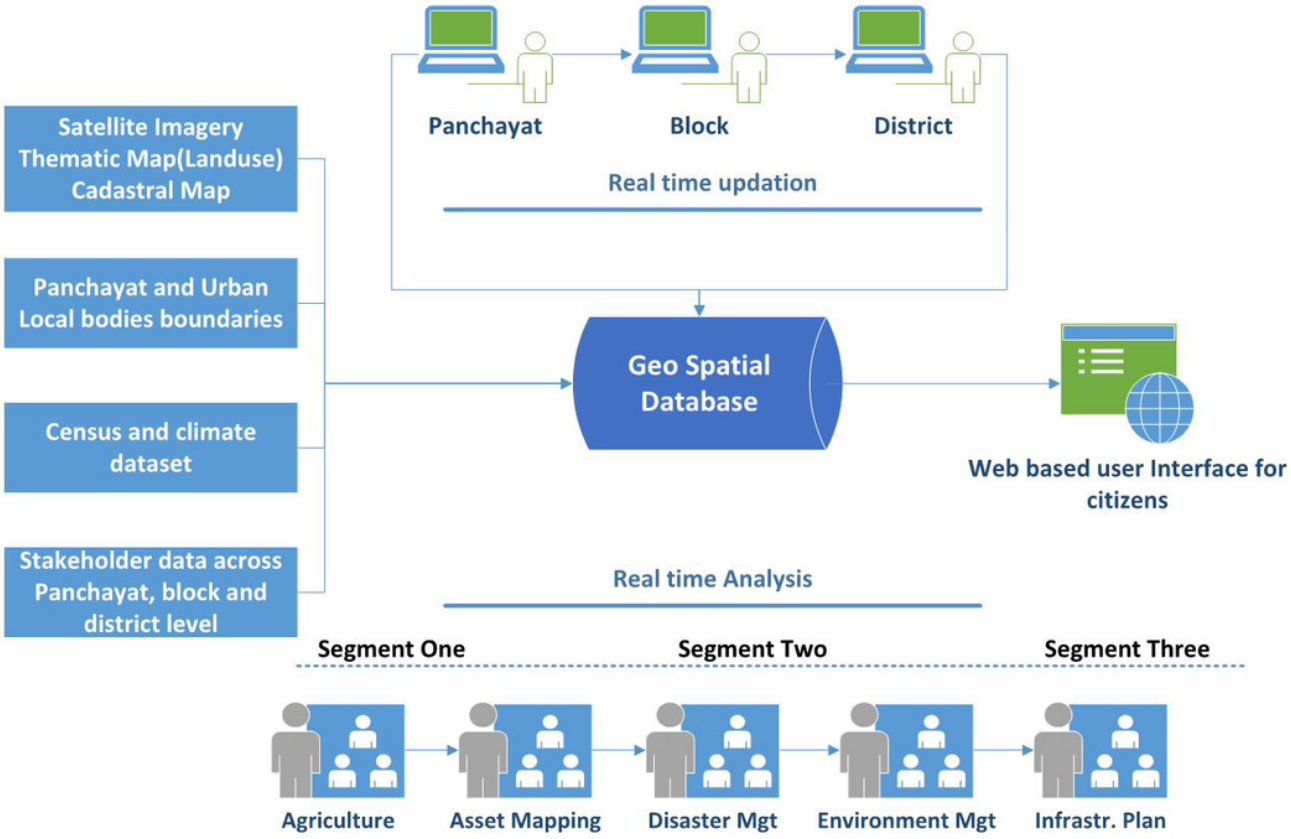


Figure 14

Web based common Geospatial database management system